

Subwoofer for Cars

KARUNESH SHUKLA

This system needs to be attached to an existing car stereo amplifier to add extra 'boom effect' to the music. It has a dedicated loudspeaker with a power amplifier to boost low frequencies that are normally omnidirectional.

The power amplifier used is an integrated class-B output amplifier in a 17-lead single-in-line (SIL) power

package. It contains 4×12W single-ended (SE) or 2×24W bridge-tied load (BTL) amplifiers. The output speaker is connected in BTL configuration in this system for better quality.

Circuit and working

Fig. 1 shows the circuit diagram of subwoofer for cars. It is built around low-noise JFET-input operational amplifiers

TL072 (IC1), car radio power amplifier TDA8561Q (IC2) and transistors BC547 (T1 and T2). IC2 has two 24W BTL output channels, one of which is connected to loudspeaker LS1. BTL is an output configuration for audio amplifiers, a form of impedance bridging where two channels of a stereo amplifier are fed the same audio signal, with one channel's electrical polarity reversed. Such an arrangement can double the voltage swing at the load.

Both channels from the car stereo are fed to the system through connectors RCA1 and RCA2. Audio from both the channels is mixed after level control. After mixing, the signals are fed to the buffer stage implemented with op-amp A1 of IC1 and the output phase can be reversed using switch S1A and S1B. This control can be useful to allow the subwoofer to be in phase with the loudspeakers of the existing car radio.

A 12dB octave variable frequency low-pass filter (LPF) built around op-amp A2 of IC1, transistor T1 and related components allows to adjust precisely (using dual-gang potmeter VR2) the low-pass frequencies between 70Hz and 150Hz. Resistor R17, capacitor C9 and transistor T2 form a simple DC voltage stabiliser for the input and filter circuitry. The filtered output is fed to inputs INV3 and -IN4 of power amplifier IC2, and the amplified output can be heard at LS1. The power supply of the circuit is 14.4V obtained from car battery. The logarithmic potmeter VR1 is used to control volume level.

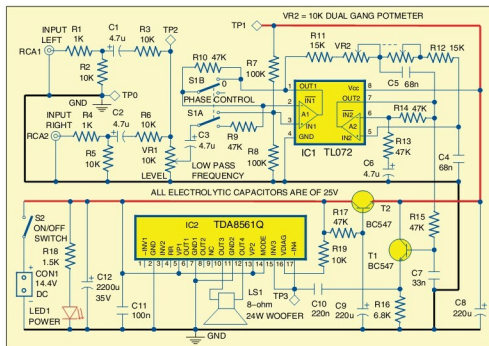


Fig. 1: Circuit diagram of the subwoofer for cars

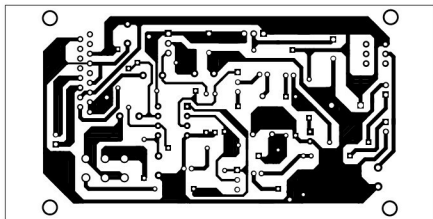


Fig. 2: An actual-size, single-side PCB for the subwoofer

Test Points

Test point	Details
TP0	GND
TP1	14.4V
TP2	Mixed audio input
TP3	Input to IC2